

Introduction to Vectors

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1. Vectors and Linear Combinations

1.1. Summary

- $3\vec{v} + 5\vec{w}$ is a typical **Linear Combination** $c\vec{v} + d\vec{w}$ of the Vectors $\vec{v}, \vec{w}$.
- For Vectors : $\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \vec{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ that combination is

$$3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 18 \end{bmatrix}$$

- The combination

$$c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

fill the whole plane, They can produce every

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

- The combination

$$c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

fill the whole plane in xyz space

- For the Equation :

$$\begin{cases} c + 2d = 1 \\ c + 3d = 0 \\ c + 4d = 0 \end{cases}$$

has no solution, because its right side vector

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

is not on that plane.

1.2. Column Vectors.

Definition : If the vector like this :

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

we call this a **Column Vector**

1.3. Linear Combination

We combine addition with scalar multiplication to produce a “**Linear Combination**” of $\vec{v}$ and $\vec{w}$:

$$c\vec{v} + d\vec{w}$$

$$c, v \in \mathbb{R}$$

here are some special Linear Combinations :

- $1\vec{v} + 1\vec{w} =$ sum of vectors in Figure 1.1a
- $1\vec{v} - 1\vec{w} =$ difference of vectors in Figure 1.1b
- $0\vec{v} + 0\vec{w} =$ **zero vector**
- $c\vec{v} + 0\vec{w} =$ vector $c\vec{v}$ in the direction of $\vec{v}$

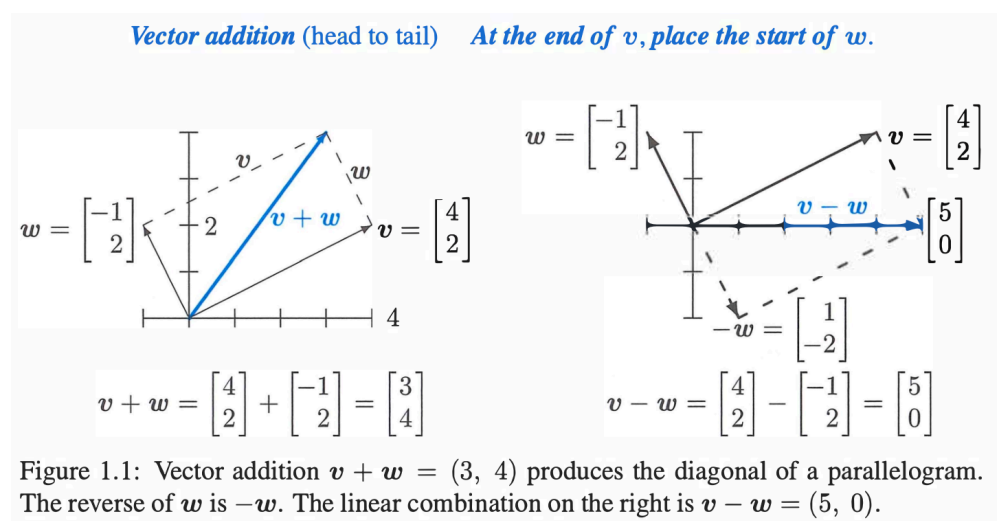


Figure 1.1: Vector addition $\vec{v} + \vec{w} = (3, 4)$ produces the diagonal of a parallelogram. The reverse of $\vec{w}$ is $-\vec{w}$. The linear combination on the right is $\vec{v} - \vec{w} = (5, 0)$.

With the **zero vector** : Any Vector Space always contains the zero vector.

1.3.1. Addition

suppose we have two vectors like this :

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}; \vec{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

then, The addition for them is

$$\vec{a} + \vec{b} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}$$

1.3.2. Subtraction

For the minus vector :

$$-\vec{v} = \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix}$$

if we plus another vector :

$$\vec{v} - \vec{v} = 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

thus, we get the **Zero Vector**. (In linear Algebra, the zero is not a number, it's a vector.)

1.3.3. The zero vector

A zero vector exists for a vector space of any dimension. If we have two vector like this :

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}; -\vec{v} = -\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

then, we adds them to produce the zero vector :

$$\vec{0} = \vec{v} - \vec{v} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

1.4. Lengths and Dot Products

1.4.1. Dot product

For two vectors :

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}; \vec{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

. the **Product** (or **inner product**) to them is

$$\vec{v} \cdot \vec{n} = v_1 \cdot n_1 + v_2 \cdot n_2$$

- The product $\vec{v}_1 \cdot \vec{v}_2 = \vec{v}_2 \cdot \vec{v}_1$, the order of $\vec{v}_1$ and $\vec{v}_2$ is not difference.

1.4.2. Perpendicular Vectors

Assume we have two vectors :

$$\vec{v}_1 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}; \vec{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

then, we can multiply the vector $\vec{v}_1$ by another vector $\vec{v}_2$ to produce :

$$\vec{v}_1 \cdot \vec{v}_2 = -4 + 4 = 0$$

,Dot Product is zero, two vectors are perpendicular. The angle between is 90° .

- Reason :

$$\begin{aligned}\cos \theta &= \frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\| \cdot \|\vec{v}_2\|} = \frac{0}{\sqrt{20} \cdot \sqrt{5}} = 0 \\ \therefore 0 &\leq \theta \leq \pi \\ \therefore \theta &= \frac{\pi}{2}\end{aligned}$$

therefore the angle between two vectors is 90° .

- Another View :

$$\vec{v}_1 - \vec{v}_2 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\begin{aligned}\|\vec{v}_1\|^2 + \|\vec{v}_2\|^2 &= 16 + 4 + 1 + 4 = 25 \\ &= \|\vec{v}_1 - \vec{v}_2\|^2\end{aligned}$$

. this is the **Pythagoras Law** ($a^2 + b^2 = c^2$). therefore the angle between two vectors is 90° .

- Notes : **The zero vector $\vec{v} = 0$ is perpendicular to every vector $\vec{w}$, because**

$$0 \cdot \vec{w} = 0$$

is always right.

1.4.3. The Lengths of the Vector

For the vector :

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

. The length of it is :

$$\|\vec{v}\| = \sqrt{(\vec{v})^2} = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + v_2^2}$$

- The angle between $\vec{v}$ and itself is not 90° , but, 0° . The reason is obviously :

$$\begin{aligned}\therefore \vec{v} \cdot \vec{v} &= \|\vec{v}\|^2 \\ \therefore \cos \theta &= \frac{\vec{v} \cdot \vec{v}}{\|\vec{v}\| \cdot \|\vec{v}\|} = \frac{\|\vec{v}\|^2}{\|\vec{v}\|^2} = 1 \Rightarrow \theta = 0^\circ\end{aligned}$$

1.4.4. The Unit Vector.

Definition : A Unit vector $\vec{u}$ is a vector whose length equals one :

$$\|\vec{u}\| = 1$$

- How to get the Unit Vector ?

Suppose we have a vector like this :

$$\vec{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

,we can produce the Unit Vector $\vec{u}$ by :

$$\vec{u} = \frac{\vec{n}}{\|\vec{n}\|} = \frac{\vec{n}}{\sqrt{n_1^2 + n_2^2}} = \begin{bmatrix} \frac{n_1}{\sqrt{n_1^2 + n_2^2}} \\ \frac{n_2}{\sqrt{n_1^2 + n_2^2}} \end{bmatrix}$$

check out :

$$\|\vec{u}\| = \sqrt{\frac{n_1^2}{n_1^2 + n_2^2} + \frac{n_2^2}{n_1^2 + n_2^2}} = 1$$

The direction of the unit vector $\vec{u}$ is same as the original vector $\vec{n}$.

- In the xy plane, the Unit Vector makes an angle “theta”

with the x axis is $(\cos \theta, \sin \theta)$:

$$\vec{u} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

. When the $\theta = 0^\circ$ the Unit vector is

$$\vec{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

that is the x axis.

when the $\theta = 90^\circ$ this is the y axis.

$$\vec{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Due to the

$$\sin^2 \theta + \cos^2 \theta = 1$$

, therefore These vectors reach out to the unit circle (with radius is 1):

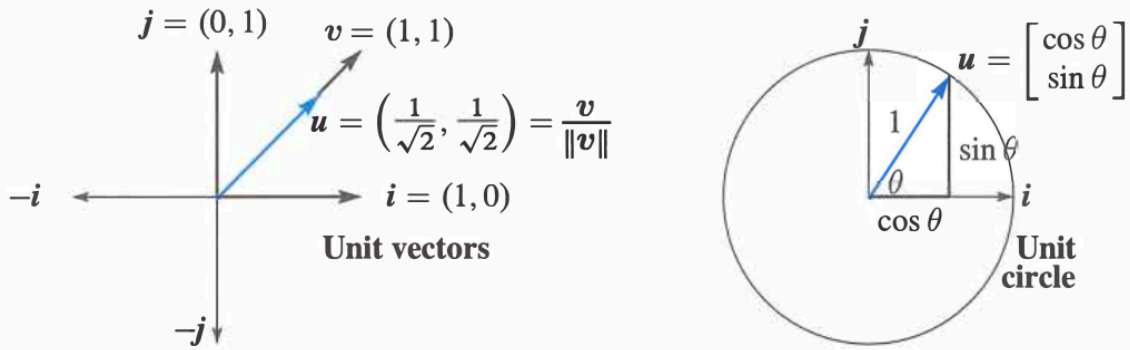


Figure 1.7: The coordinate vectors i and j . The unit vector u at angle 45° (left) divides $v = (1, 1)$ by its length $\|v\| = \sqrt{2}$. The unit vector $u = (\cos \theta, \sin \theta)$ is at angle θ .

1.4.5. The angle between two vectors

If the vectors

$$\vec{v} \text{ and } \vec{w}$$

are nonzero vectors, then, the angle between them is :

$$\theta = \arccos\left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \cdot \|\vec{w}\|}\right)$$

- For unit vectors $\vec{u}$ and $\vec{U}$, the cosine of them is :

$$\cos \theta = \frac{\vec{u} \cdot \vec{U}}{1} = \vec{u} \cdot \vec{U}$$

therefore,

$$|\vec{u} \cdot \vec{U}| < 1$$

- Due to the

$$-1 \leq \cos \theta \leq 1 \Rightarrow |\cos \theta| \leq 1$$

$$\Rightarrow \frac{|\vec{v} \cdot \vec{w}|}{\|\vec{v}\| \cdot \|\vec{w}\|} \leq 1$$

so, we have

$$\text{Schwarz Inequality : } |\vec{v} \cdot \vec{w}| \leq \|\vec{v}\| \cdot \|\vec{w}\|$$

$$\text{Triangle Inequality : } \|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$$

1.5. Geometric Mean for Vectors

For two vectors :

$$\vec{v} = \begin{bmatrix} a \\ b \end{bmatrix}; \vec{w} = \begin{bmatrix} b \\ a \end{bmatrix}$$

the length of this vector is

$$\|\vec{v}\| = \sqrt{a^2 + b^2}$$

$$\|\vec{w}\| = \sqrt{a^2 + b^2}$$

. The Schwarz Inequality tells our :

$$\begin{aligned} |\vec{v} \cdot \vec{w}| &= |ab + ba| \\ &= 2ab \leq \|\vec{v}\| \cdot \|\vec{w}\| = a^2 + b^2 \\ &\Rightarrow a^2 + b^2 \geq 2ab \end{aligned}$$

. If

$$x = a^2$$

$$y = b^2$$

then, we get the “Arithmetic Mean “:

$$\frac{x + y}{2} \geq \sqrt{xy}$$

so ,

$$\frac{a^2 + b^2}{2} \geq ab \Leftrightarrow \frac{x + y}{2} \geq \sqrt{xy}$$

“the Arithmetic mean is large than the Geometric mean “

2. Matrices

2.1. The Difference Matrix

Suppose we have three vectors :

$$\vec{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}; \vec{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}; \vec{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

we combine these by the “Matrices”:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

- Matrix times the vector : **combination of columns of the A**

$$A\vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = \vec{b}$$

this matrix A acts on the vector $\vec{x}$, the output is the combination of the columns of A .

Simultaneously, the matrix A is called “**Difference Matrix**”. (because the $\vec{b}$ contains the differences of the input vector $\vec{x}$)

- The Multiplication for matrix :

Multiplication a row at a time :

$$A\vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} (1, 0, 0) \cdot (x_1, x_2, x_3) \\ (-1, 1, 0) \cdot (x_1, x_2, x_3) \\ (0, -1, 1) \cdot (x_1, x_2, x_3) \end{bmatrix}$$

So, $A\vec{x}$ is also dot products with rows.

2.2. Linear Equations

- The Old Question is : Compute the linear combination $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{b}$.
- The New Question is : Which combination of the vectors

$$\vec{v}, \vec{u}, \vec{w}$$

produces the particular vector $\vec{b}$? This is a **Inverse Problem**- to find the input $\vec{x}$ to produce the particular vector $\vec{b}$.

Fortunately, due to the Equations :

$$\begin{cases} x_1 = b_1 \\ -x_1 + x_2 = b_2 \\ -x_2 + x_3 = b_3 \end{cases}$$

is a “triangle” Equations. So, we can solve this Equations from top to bottom. get the :

$$\begin{cases} x_1 = b_1 \\ x_2 = b_1 + b_2 \\ x_3 = b_1 + b_2 + b_3 \end{cases}$$

.

Look at a specific choices :

$$A\vec{x} = 0 = \vec{b} \Rightarrow \vec{b} = 0$$

For a Difference matrix A , if $A\vec{x} = 0$, the $\vec{x}$ must be 0. (When this time, we call the matrix A is **invertible**. we can recover $\vec{x}$ from $\vec{b}$:

$$\vec{x} = A^{-1} \cdot \vec{b}$$

)

2.3. The identity matrix

The “Identity Matrix” :

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Elements on the diagonal always is one and other place always is zero.

Any Matrix multiply by identity matrix still is itself :

$$AI = A; IA = A$$

The effect of I on another matrix is similar to that of the number 1.

2.4. The Inverse Matrix

$$A\vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \vec{b}$$

For this Equation we have the Inverse matrix that will undo the effects produced by A and we call “inverse Matrix” A^{-1} :

$$A^{-1}A\vec{x} = A^{-1}\vec{b} = \vec{x}$$

therefore ,

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ \Rightarrow A^{-1}A\vec{x} &= I\vec{x} = \vec{b} \\ \Rightarrow \vec{x} &= A^{-1}\vec{b} \\ \therefore A^{-1} &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

- The connection between inverse matrices and integrals :

The x change to the function $x(t)$, the matrix A change to the $\frac{d}{dt}x(t)$ then, we get :

$$Ax = b \text{ and } x = A^{-1}b \Leftrightarrow \frac{dx}{dt} = b \text{ and } x(t) = \int_0^x x(t)dt$$

The differences of squares 0, 1, 4, 9 are odd numbers 1, 3, 5. The derivative of $x(t) = t^2$ is $2t$. A perfect analogy would have produced the even numbers $b = 2, 4, 6$ at times $t = 1, 2, 3$. But differences are not the same as derivatives, and our matrix A produces not $2t$ but $2t - 1$:

Backward $x(t) - x(t - 1) = t^2 - (t - 1)^2 = t^2 - (t^2 - 2t + 1) = 2t - 1.$ (9)

The Problem Set will follow up to show that “forward differences” produce $2t + 1$. The best choice (not always seen in calculus courses) is a **centered difference** that uses $x(t + 1) - x(t - 1)$. Divide that Δx by the distance Δt from $t - 1$ to $t + 1$, which is 2:

Centered difference of $x(t) = t^2$ $\frac{(t + 1)^2 - (t - 1)^2}{2} = 2t$ exactly. (10)

Difference matrices are great. Centered is the best. Our second example is *not invertible*.

2.5. The Cyclic Matrix

$$C = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Suppose we have below euqation :

$$C\vec{x} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = \vec{b}$$

For this Equation, it's impossible to find the solution to $Cx = b$, as the three euqations either have infinitely many solutions (sometime) or else no solution(usually).

- $Cx = b$ have infinitely solutions :

$$\begin{bmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is solved by all vectors :

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c \\ c \\ c \end{bmatrix}$$

the underdetermind constant c can be any number.

- $Cx = b$ no solution :

$$\begin{bmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

This equation has no solution, because its left sides add to zero, but right sides add to 9. In other view :

All combinations

$$x_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

lie on the same plane given by

$$b_1 + b_2 + b_3 = 0$$

and the vector $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ is not in this plane. so we cannot produce the vector $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ by combine those vectors.

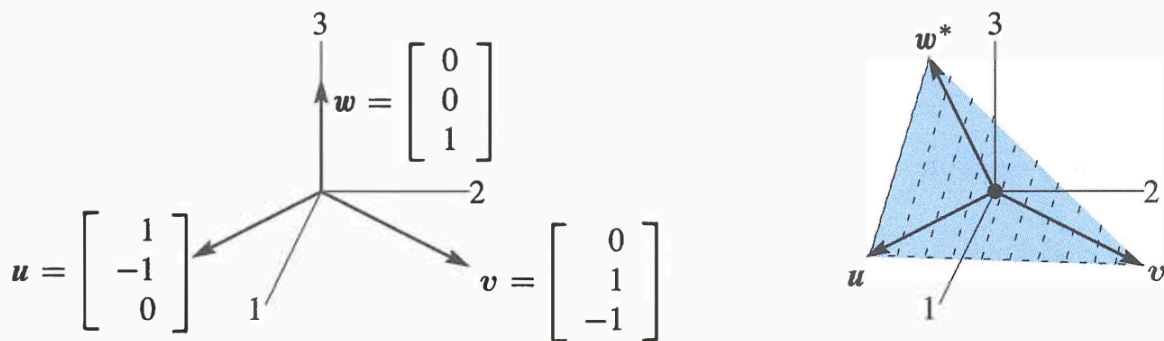


Figure 1.10: Independent vectors u, v, w . Dependent vectors u, v, w^* in a plane.

The combinations of Independent vectors can fill the whole vector space.

$$x_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

2.6. Independence and Dependence

Suppose we have two vectors

$$\vec{v}; \vec{u}$$

. The key Question is whether the third vector is in that plane (constructed by $\vec{u} + \vec{v}$)

- **Independence** : If the third vector $\vec{w}$ is not on that plane.
- **Dependence** : If the third vector in that plane :

$$\vec{v} + \vec{u} = \vec{w}$$

In another view :

- **Independence** : For the equation $Ax = 0$ only one solution A is **invertible**.
- **Dependence** : For the equation $Cx = 0$ has many solutions, C is **singular**.

So, we have below conclusions :

1. If a matrix A is invertible, the columns of A must be **Independent**. and $Ax = 0$ *Only one solution* ($x = 0$).
2. If a matrix C is singular, the columns of C must be **Dependent**. and $Cx = 0$ *has many nonzero solutions*.